$$= \lim_{x \to \frac{\pi}{4}} \frac{2\cos\frac{\pi}{4} \cdot \sin\left(\frac{\pi}{4} - x\right)}{x - \frac{\pi}{4}}$$

$$= \lim_{x \to \frac{\pi}{4}} \frac{-2\cos\frac{\pi}{4} \cdot \sin\left(x - \frac{\pi}{4}\right)}{x - \frac{\pi}{4}}$$

$$= -2\left(\frac{1}{\sqrt{2}}\right) \cdot 1$$

$$= -\sqrt{2}$$ #.

19. $\lim_{x \to 1} \frac{1+\cos \pi x}{\tan^2 \pi x}$

$\rightarrow$ The function takes $\frac{0}{0}$ form when $x=1$

$= \lim_{x \to 1} \frac{1+\cos \pi x}{\tan^2 \pi x} \times \frac{1-\cos \pi x}{1-\cos \pi x}$

$= \lim_{x \to 1} \frac{1^2-(\cos \pi x)^2}{\tan^2 \pi x} \cdot \lim_{x \to 1} \frac{1}{1-\cos \pi x}$

$= \lim_{x \to 1} \frac{1-\cos^2 \pi x}{\tan^2 \pi x} \cdot \frac{1}{1-\cos \pi}$

$= \lim_{x \to 1} \frac{\cancel{\sin^2 \pi x} \times \cos^2 \pi x}{\cancel{\sin^2 \pi x}} \cdot \frac{1}{1-(-1)}$

$= \cos^2 \pi \cdot \frac{1}{2}$

$= 1 \times \frac{1}{2}$

$= \frac{1}{2}$ Ans.

20. $\lim_{x \to y} \frac{\tan x - \tan y}{x-y}$

$\rightarrow$ The function takes $\frac{0}{0}$ form when $x=y$

$= \lim_{x \to y} \frac{1}{x-y} \left(\frac{\sin x}{\cos x} - \frac{\sin y}{\cos y} \right)$

$= \lim_{x \to y} \frac{1}{x-y} \left(\frac{\sin x . \cos y - \cos x . \sin y}{\cos x . \cos y} \right)$

$= \lim_{x \to y} \frac{1}{x-y} \left(\frac{\sin (x-y)}{\cos x . \cos y} \right)$

$= \lim_{x \to y} \frac{\sin (x-y)}{x-y} \times \frac{1}{\cos x . \cos y}$

$= 1 \times \dfrac{1}{\cos y \cdot \cos y}$

$= \dfrac{1}{\cos^2 y}$

$= \sec^2 y$

21. $\lim\limits_{x \to y} \dfrac{\cos x - \cos y}{x - y}$

→ The function takes $\frac{0}{0}$ form when $x = y$

$= \lim\limits_{x \to y} \dfrac{-2\sin\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{x-y}{2}\right)}{x-y}$

$= \lim\limits_{x \to y} \dfrac{-\cancel{2} \sin\left(\frac{x-y}{2}\right)}{\frac{x-y}{2}} \times \dfrac{1}{\cancel{2}} \times \sin\left(\frac{x+y}{2}\right)$

$= \cancel{\lim} \; -1 \times \sin\left(\frac{y+y}{2}\right)$

$= -\sin \dfrac{\cancel{2}y}{\cancel{2}}$

$= -\sin y$ //.

22. $\lim\limits_{x \to a} \dfrac{\sin x - \sin a}{\sqrt{x} - \sqrt{a}}$

→ The function takes $\frac{0}{0}$ form when $x = a$

$= \lim\limits_{x \to a} \dfrac{\sin x - \sin a}{\sqrt{x} - \sqrt{a}} \times \dfrac{\sqrt{x} + \sqrt{a}}{\sqrt{x} + \sqrt{a}}$

$$= \lim_{x \to a} \frac{\sin x - \sin a}{(\sqrt{x})^2 - (\sqrt{a})^2} \times \sqrt{x} + \sqrt{a}$$

$$= \lim_{x \to a} \frac{2\cos\left(\frac{x+a}{2}\right) \cdot \sin\left(\frac{x-a}{2}\right)}{x-a} \cdot \lim_{x \to a} \sqrt{x} + \sqrt{a}$$

$$= \lim_{x \to a} \frac{\cancel{2} \sin\left(\frac{x-a}{2}\right)}{\frac{x-a}{2}} \times \frac{1}{\cancel{2}} \times \cos\left(\frac{x+a}{2}\right) \times (\sqrt{a} + \sqrt{a})$$

$$= \lim_{x \to a} 1 \times \cos\left(\frac{x+a}{2}\right) \times 2\sqrt{a}$$

$$= 1 \times \cos\left(\frac{\cancel{2}a}{\cancel{2}}\right) \times 2\sqrt{a}$$

$$= 2\sqrt{a} \cos a \#.$$

23. $\lim_{x \to \theta} \dfrac{x \sin\theta - \theta \sin x}{x - \theta}$

→ The function takes $\frac{0}{0}$ form when $x = \theta$

$$= \lim_{x \to \theta} \frac{1}{x-\theta} (x \sin\theta - x \sin x + x \sin x - \theta \sin x)$$

$$= \lim_{x \to \theta} \frac{x(\sin\theta - \sin x) + \sin x (x - \theta)}{x - \theta}$$

$$= \lim_{x \to \theta} \frac{x(\sin\theta - \sin x)}{x - \theta} + \lim_{x \to \theta} \frac{\cancel{(x-\theta)} \cdot \sin x}{\cancel{x-\theta}}$$

$$= \lim_{x \to \theta} \frac{x\left[2\cos\left(\frac{\theta + x}{2}\right) \cdot \sin\left(\frac{\theta - x}{2}\right)\right]}{x - \theta} + \sin\theta$$

$$= \sin\theta + \lim_{x \to \theta} -\cancel{2}x \cdot \frac{\sin\left(\frac{x-\theta}{2}\right)}{\frac{x-\theta}{2}} \times \frac{1}{\cancel{2}} \times \cos\left(\frac{\theta + \theta}{2}\right)$$

$= \sin\theta - \theta \cdot 1 \cdot \cos\theta$

$= \sin\theta - \theta\cos\theta$ Ans.

24. $\lim_{x \to \theta} \dfrac{x\cot\theta - \theta\cot x}{x - \theta}$

→ The function takes $\frac{0}{0}$ form when $x = \theta$

$= \lim_{x \to \theta} \dfrac{1}{x-\theta} \left[x\cot\theta - x\cot x + x\cot x - \theta\cot x \right]$

$= \lim_{x \to \theta} \dfrac{1}{x-\theta} \left[x(\cot\theta - \cot x) + \cot x(x - \theta) \right]$

$= \lim_{x \to \theta} \dfrac{\cot x \cancel{(x-\theta)}}{\cancel{x-\theta}} + \lim_{x \to \theta} \dfrac{x(\cot\theta - \cot x)}{(x-\theta)}$

$= \cot\theta + \lim_{x \to \theta} \dfrac{\theta}{(x-\theta)} \left[\dfrac{\cos\theta}{\sin\theta} - \dfrac{\cos x}{\sin x} \right]$

$= \cot\theta + \lim_{x \to \theta} \dfrac{\theta}{x-\theta} \left[\dfrac{\sin x \cdot \cos\theta - \cos x \cdot \sin\theta}{\sin\theta \cdot \sin x} \right]$

$= \cot\theta + \lim_{x \to \theta} \dfrac{\theta}{x-\theta} \dfrac{\sin(x-\theta)}{\sin\theta \cdot \sin x}$

$= \cot\theta + \lim_{x \to \theta} \dfrac{\sin(x-\theta)}{x-\theta} \cdot \dfrac{\theta}{\sin\theta \cdot \sin x}$

$= \cot\theta + 1 \cdot \dfrac{\theta}{\sin\theta \cdot \sin\theta}$

$= \cot\theta + \dfrac{\theta}{\sin^2\theta}$

$= \cot\theta + \theta\,\mathrm{cosec}^2\theta$ Ans.

25. $\lim_{x \to \theta} \dfrac{x \tan x - \theta \tan\theta}{x - \theta}$

→ The function takes $\frac{0}{0}$ form when $x = \theta$

$= \lim_{x \to \theta} \dfrac{x \tan x - x \tan\theta + x\tan\theta - \theta\tan\theta}{x-\theta}$

$= \lim_{x \to \theta} \dfrac{(\tan x - \tan\theta)\, x + \cancel{\tan}\,(x-\theta)\tan\theta}{x-\theta}$

$= \lim_{x \to \theta} \dfrac{(x-\theta)\tan\theta}{(x-\theta)} + \dfrac{x(\tan x - \tan\theta)}{x-\theta}$

$= \tan\theta + \lim_{x \to \theta} \dfrac{x}{x-\theta}\left[\dfrac{\sin x}{\cos x} - \dfrac{\sin\theta}{\cos\theta}\right]$

$= \tan\theta + \lim_{x \to \theta} \dfrac{x}{x-\theta}\left[\dfrac{\sin x.\cos\theta - \cos x.\sin\theta}{\cos x.\cos\theta}\right]$

$= \tan\theta + \lim_{x \to \theta} \dfrac{x}{x-\theta} \; \dfrac{\sin(x-\theta)}{\cos x.\cos\theta}$

$= \tan\theta + \lim_{x \to \theta} \dfrac{\cancel{\theta}\, x}{\cos\theta\cdot\cos\theta} \times \dfrac{\sin(x-\theta)}{x-\theta}$

$= \tan\theta + \dfrac{\theta}{\cos^2\theta} \times 1$

$= \tan\theta + \theta\sec^2\theta$

26. $\lim_{y \to x} \dfrac{y\sec y - x\sec x}{y-x}$

→ The function takes $\frac{0}{0}$ form when $y = x$

$= \lim_{y \to x} \dfrac{1}{y-x}\left[y\sec y - y\sec x + y\sec x - x\sec x\right]$

$= \lim_{y \to x} \dfrac{1}{y-x}\left[y(\sec y - \sec x) + (y-x)\sec x\right]$

$$= \lim_{y\to x} \frac{1}{y-x}[y(\sec y - \sec x)] + \lim_{y\to x} \frac{\cancel{(y-x)}\sec x}{\cancel{(y-x)}}$$

$$= \lim_{y\to x} \frac{y}{y-x}\left[\frac{1}{\cos y} - \frac{1}{\cos x}\right] + \sec x$$

$$= \lim_{y-x} \frac{y}{y-x}\left[\frac{\cos x - \cos y}{\cos x . \cos y}\right] + \sec x$$

$$= \lim_{y\to x} \frac{y}{y-x}\frac{\left[2\sin\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{y-x}{2}\right)\right]}{\cos x . \cos y} + \sec x$$

$$= \lim_{y\to x} \frac{2y}{y-x} \times \frac{\left[\sin\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{y-x}{2}\right)\right]}{\cos x . \cos y} + \sec x$$

$$= \lim_{y\to x} \frac{\sin\left(\frac{y-x}{2}\right)}{\frac{y-x}{2}} \times \frac{1}{\cancel{2}} \times \cancel{2}y \times \frac{\sin\left(\frac{x+y}{2}\right)}{\cos x . \cos x} + \sec x$$

$$= 1 \times x * \frac{\sin x}{\cos^2 x} + \sec x$$

$$= \sec x + x \sec x . \tan x$$

$$= \sec x(1 + x\tan x)$$ Ans.

21. Evalute the following limits

27 (a) $\lim_{x\to 0} \frac{e^{5x}-1}{x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$$= \lim_{x\to 0} \frac{e^{5x}-1}{5x} \times 5$$

$= 1 \times 5$

$= 5$ Ans.

(28) b) $\lim_{x \to 0} \frac{e^{6x} - e^{7x}}{x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$= \lim_{x \to 0} \frac{e^{6x} - 1 - e^{7x} + 1}{x}$

$= \lim_{x \to 0} \frac{e^{6x} - 1 - (e^{7x} - 1)}{x}$

$= \lim_{x \to 0} \frac{e^{6x} - 1}{x} - \frac{e^{7x} - 1}{x}$

$= \lim_{x \to 0} \frac{e^{6x} - 1}{6x} \times 6 - \frac{e^{7x} - 1}{7x} \times 7$

$= 1 \times 6 - 1 \times 7$

$= 6 - 7$

$= -1$ Ans.

(29) c) $\lim_{x \to 0} \frac{e^{5x} - 1}{x \cdot 3^{x}}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$= \lim_{x \to 0} \frac{e^{5x} - 1}{x} \times \lim_{x \to 0} \frac{1}{3^{x}}$

$= \lim_{x \to 0} \frac{e^{5x} - 1}{5x} \times 5 \times \frac{1}{3^{0}}$

$= \cancel{\lim}\ 1 \times 5 \times \frac{1}{1}$

$= 5$ Ans.

30 d) $\lim_{x\to 0} \frac{a^{3x}-a^{5x}}{x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$= \lim_{x\to 0} \frac{a^{3x}-1-a^{5x}+1}{x}$

$= \lim_{x\to 0} \frac{a^{3x}-1-(a^{5x}-1)}{x}$

$= \lim_{x\to 0} \frac{a^{3x}-1}{x} - \frac{a^{5x}-1}{x}$

$= \lim_{x\to 0} \frac{a^{3x}-1}{3x}\times 3 - \frac{a^{5x}-1}{5x}\times 5$

$= \ln a \times 3 - \ln a \times 5$

$= \cancel{\ln 3a - \ln 5a}$

$= \ln (3-5)$ in a

$= \ln -2$ in a #.

31 e) $\lim_{x\to 0} \frac{a^{x}-b^{x}}{x}$

→ The function takes $\frac{0}{0}$ form when $x=0$

$= \lim_{x\to 0} \frac{a^{x}-1-b^{x}+1}{x}$

$= \lim_{x\to 0} \frac{a^{x}-1-(b^{x}-1)}{x}$

$= \lim_{x\to 0} \frac{a^{x}-1}{x} - \frac{b^{x}-1}{x}$

$= \ln a - \ln b$

$= \ln\left(\frac{a}{b}\right)$ #.

32 f) $\lim_{x \to 1} \frac{\ln x}{x-1}$

→ The function takes indeterminate form when $x = 1$

$= \lim_{x \to 1} \frac{\ln x}{x-1}$

let, $x-1 = y \Rightarrow x = y+1$

$= \lim_{x \to 1} \frac{\ln(y+1)}{y}$

$= \lim_{y \to 0} \frac{\ln(y+1)}{y}$

$= 1$ #.

33 g) $\lim_{x \to 2} \frac{x-2}{\ln(x-1)}$

→ The function takes indeterminate form when $x = 2$

→ let, $x-2 = y \Rightarrow x = y+2$

$= \lim_{x \to 2} \frac{y}{\ln(y+2-1)}$

$= \cancel{\lim_{x \to 2} \frac{y}{\ln(y+1)}}$

$= \lim_{y \to 0} \frac{y}{\ln(y+1)}$

$= 1$

#.

One Sided Limits

1. Find the limits of the following functions at the specified points.

1 a) $\lim_{x \to 0} |x|$

$= |0|$

$= 0$ #.

2 b) $\lim_{x \to 3} \frac{|x-3|}{x-3}$

$= \frac{|3-3|}{3-3}$

$= \frac{0}{0}$ (indeterminate)

$\therefore$ limit doesn't exist

3 c) $\lim_{x \to 2} |x-2|$

$= |2-2|$

$= |0|$

$= 0$

4 d) $\lim_{x \to 2} \frac{x^2-4}{x-2}$

$= \frac{2^2-4}{2-2}$

$= \frac{4-4}{0}$

$= \frac{0}{0}$ (indeterminate)

#

$= \lim_{x \to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}}$

$= 2+2$

$= 4$ #.

2. Evaluate the following limits, if exists.

5 a) $f(x) = \begin{cases} x^2 - 1 & \text{for } x < 1 \\ x^2 + 1 & \text{for } x \geq 1 \end{cases}$ at $x = 1$

→ L.H.L

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} x^2 - 1$$

$$= 1^2 - 1$$

$$= 1 - 1$$

$$= 0$$

R.H.L

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} x^2 + 1$$

$$= 1^2 + 1$$

$$= 1 + 1$$

$$= 2$$

Since L.H.L and R.H.L of the function are not equal so, limit of the function doesn't exist.

6 b) $f(x) = \begin{cases} 4x - 2 & \text{for } x \geq 2 \\ 2x & \text{for } x < 2 \end{cases}$ at $x = 2$

→ L.H.L

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} 2x$$

$$= 2 \times 2$$

$$= 4$$

R.H.L

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} 4x-2$$

$= 4\times 2-2$

$= 8-2$

$= 6$

Since L.H.L and R.H.L of the function are not equal so, limit of the function doesnot exist.

7 c) $f(x) = \begin{cases} x+2 & \text{when } x \geq 0 \\ 4x+2 & \text{when } x < 2 \end{cases}$ at $x=0$

→ L.H.L

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} x+2$$

$= 0+2$

$= 2$

R.H.L

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} 4x+2$$

$= 4\times 0+2$

$= 0+2$

$= 2$

Since, L.H.L and R.H.L of the function are equal so, limit of the function is 2.

8 d) $f(x) = \begin{cases} 3x^2+2 & \text{if } x<1 \\ 2x+3 & \text{if } x \geq 1 \end{cases}$ at $x=1$

→ L.H.L

$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} 3x^2+2$

$= 3\times 1^2 + 2$

$= 3+2$

$= 5$

R.H.L

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} 2x+3$

$= 2\times 1 + 3$

$= 2+3$

$= 5$

Since, L.H.L = R.H.L = 5

∴ limit of the function is 5.

3. Find the value of K if the limit of the functions exist.

9 a) $f(x) = \begin{cases} Kx+5 & \text{for } x \leq 2 \\ x-1 & \text{for } x>2 \end{cases}$ exist at $x=2$

→ Soln,

~~f(2) =~~ R.H.L

$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} x-1$

$= 2-1$

$= 1$

Now,

L.H.L

$$\lim_{x\to 2^-} = f(2) = \lim_{x\to 2^-} kx+5$$

$$1 = k\times 2+5$$

or, $1 = 2k+5$

or, $2k = 1-5$

or, $2k = -4$

or, $k = -\frac{4}{2}$

or, $k = -2$

$\therefore k = -2$ Ans.

10) f(x) = $\begin{cases} \frac{2x^2-18}{x-3} & \text{for } x>3 \\ k-4 & \text{for } x\leq 3 \end{cases}$ exist at $x=3$

→ Soln,

R.H.L

$$\lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} \frac{2x^2-18}{x-3}$$

$$= \lim_{x\to 3^+} \frac{2(x^2-9)}{x-3}$$

$$= \lim_{x\to 3^+} \frac{2(x+3)\cancel{(x-3)}}{\cancel{(x-3)}}$$

$$= 2(3+3)$$

$$= 2\times 6$$

$$= 12$$

L.H.L

$$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} k-4$$

$$12 = k-4$$

or, $k = 12 + 4$

or, $k = 16$

$\therefore k = 16$ #

11 c) $f(x) = \begin{cases} \dfrac{x^2-25}{x-5} & \text{for } x>5 \\ 10 & \text{for } x=5 \\ 2k-4 & \text{for } x<5 \end{cases}$ exist at $x=5$

→ Soln,

$f(5) = 10$

R.H.L

$\lim_{x\to 5^+} f(x) = \lim_{x\to 5^+} \dfrac{x^2-25}{x-5}$

$= \lim_{x\to 5^+} \dfrac{(x+5)\cancel{(x-5)}}{\cancel{(x-5)}}$

$= 5+5$

$= 10$

L.H.L

$\lim_{x\to 5^-} f(x) = \lim_{x\to 5^-} 2k-4$

or, $10 = 2\times k - 4$

or, $2k = 10 + 4$

or, $k = \dfrac{14}{2}$

or, $k = 7$

$\therefore k = 7$ #.

- Continuity

Exercise 17.4

1. Test the continuity of the given functions at the specified points.

① a) $f(x) = \dfrac{x^2-9}{x-3}$ at $x=3$

→ Soln,

$f(3) = \dfrac{3^2-9}{3-3}$

$= \dfrac{0}{0}$ Which does not exists.

Hence $f(x)$ is discontinuous at $x=3$.

② b) $f(x) = \dfrac{1}{x-5}$ at $x \neq 5$

→ Soln,

$x \neq 5,\ x = 5+h,\ h \neq 0$

$f(5+h) = \dfrac{1}{5+h-5}$

$= \dfrac{1}{h}$ Which exists.

$\lim\limits_{x \to 5+h} f(x) = \lim\limits_{x \to 5+h} \dfrac{1}{x-5}$

$= \dfrac{1}{5+h-5}$

$= \dfrac{1}{h}$

Since $\lim_{x \to 5+h} f(x) = f(5+h)$ so $f(x)$ is continuous at $x \pm 5$.

3 c) $f(x) = \frac{1}{5x}$ at $x = 0$

→ Soln,

$f(0) = \frac{1}{5 \times 0}$

$= \frac{1}{0}$

$= \infty$ Which does not exists.

4 d) $f(x) = \frac{|x-4|}{x-4}$ at $x = 4$

→ Here,

Left hand limit of $f(x)$ at $x = 4$ is

L.H.L

$\lim_{x \to 4^-} f(x) = \lim_{x \to 4^-} \frac{|x-4|}{(x-4)}$

$= \lim_{x \to 4^-} \frac{-(x-4)}{x-4}$

$= -1$

and Right hand limit of $f(x)$ at $x = 4$ is

R.H.L

$\lim_{x \to 4^+} f(x) = \lim_{x \to 4^+} \frac{|x-4|}{(x-4)}$

$= \lim_{x \to 4^+} \frac{(x-4)}{(x-4)}$

$= 1$

Since L.H.L $\neq$ R.H.L so, $f(x)$ is discontinuous at $x=5$.

2. Discuss the continuity of given functions at the specified points:

5 a) $f(x) = \begin{cases} 2x^2+3 & \text{for } x \leq 1 \\ 7x-2 & \text{for } x > 1 \end{cases}$ at $x=1$

→ Soln,

$f(1) = 2\times1^2+3$

$= 2+3 = 5$

R.H.L

$\lim_{x\to2^+} f(x) = \lim_{x\to1^+} 7x-2$

$= 7\times1-2$

$= 7-2$

$= 5$

L.H.L

$\lim_{x\to2^-} f(x) = \lim_{x\to1^-} 2x^2+3$

$= 2\times1^2+3$

$= 2+3$

$= 5$

Since, L.H.L = R.H.L = 5 the limit of

a function exists.

$$\therefore \lim_{x\to 2} f(x) = 5$$

Since $\lim_{x\to 2} f(x) = f(1) = 5$, ~~The~~ Given function is Continuous.

6 b) $f(x) = \begin{cases} 4-x^2 & \text{if } x \le 0 \\ 4+x & \text{if } x > 0 \end{cases}$ at $x=0$

→ Soln,

$$f(0) = 4-0^2$$
$$= 4-0$$
$$= 4$$

L.H.L

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} 4-x^2$$
$$= 4-0^2$$
$$= 4-0$$
$$= 4$$

R.H.L

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} 4+x$$
$$= 4+0$$
$$= 4$$

Since L.H.L = R.H.L = 4 , the limit of a function exists.

$\therefore \lim_{x \to 0} f(x) = 4$

Since $\lim_{x \to 0} f(x) = f(0)$, Given function is continuous.

7 c) $f(x) = \begin{cases} \dfrac{x^2 - 25}{x - 5} & \text{for } x \neq 5 \\ 2x & \text{for } x = 5 \end{cases}$ at $x = 5$

→ Soln,

$f(5) = 2 \times 5$

$= 10$

For $x \neq 5$

$\lim_{x \to 5} f(x) = \lim_{x \to 5} \dfrac{x^2 - 25}{x - 5}$

$= \lim_{x \to 5} \dfrac{(x+5)\cancel{(x-5)}}{\cancel{(x-5)}}$

$= \lim_{x \to 5} x + 5$

$= 5 + 5$

$= 10$

Since $\lim_{x \to 5} f(x) = f(5) = 10$, $\therefore$ Given function is continuous.

d) $f(x) = \begin{cases} \dfrac{|x|}{x} & \text{for } x \neq 0 \\ x & \text{for } x = 0 \end{cases}$ at $x = 0$

→ Soln,

$f(0) = \cancel{x}\ 0$

For $x \neq 0$ When $x > 1$

R.H.L

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{|x|}{x}$$
$$= \frac{x}{x}$$
$$= 1$$

When $x < 1$

L.H.L

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \frac{|x|}{x}$$
$$= \frac{-x}{x}$$
$$= -1$$

Since L.H.L ≠ R.H.L limit of the given function doesn't exist.

∴ ~~The~~ Given function is discontinuous.

3.

9 a) Define continuity of a function at a point. Show that the function defined as follows.

$$f(x) = \begin{cases} 2x+1 & \text{for } x<1 \\ 3 & \text{for } x=1 \\ 3x & \text{for } x>1 \end{cases}$$ is continuous at $x=1$.

→ A function $f(x)$ is said to be continuous at a given point $x=a$ if the ~~func~~ following conditions are satisfied

i) The functional value of $f(x)$ at $x=a$ exists i.e. $f(a)$ exists.

ii) The left ~~hat~~ hand limit of $f(x)$ at $x=a$ exists i.e, $\lim_{x \to a^-} f(x)$ exists.

iii) The Right hand limit of $f(x)$ at $x=a$ exists i.e, $\lim_{x \to a^+} f(x)$ exists.

iv) $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$.

→ Soln,

$f(1) = 3$

L.H.L $\qquad$ L.H.L

$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} 2x+1$

$$= 2 \times 1 + 1$$
$$= 2 + 1$$
$$= 3$$

R.H.L

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} 3x$$
$$= 3 \times 1$$
$$= 3$$

Since L.H.L = R.H.L = 3, the limit of the function exists.

$$\therefore \lim_{x \to 1} f(x) = 3$$

Since $\lim_{x \to 1} f(x) = f(1)$, Given function is continuous.

10 b) A function $f(x)$ is defined as follows

$$f(x) = \begin{cases} 3 + 2x & \text{for } -3/2 \leq x < 0 \\ 3 - 2x & \text{for } 0 \leq x < 3/2 \\ -3 - 2x & \text{for } x \geq 3/2 \end{cases}$$

Show that $f(x)$ is continuous at $x = 0$ and discontinuous at $x = \frac{3}{2}$

→ Soln,

$$f(0) = 3 - 2 \times 0$$
$$= 3 - 0$$

$= 3$

L.H.L

$$\lim_{x \to \frac{3}{2}^-} f(x) = \lim_{x \to \frac{3}{2}^-} 3 + 2x$$

$$= 3 + \cancel{2} \times \frac{3}{\cancel{2}}$$

$$= 3 + 3$$

$$= 6$$

$$f\left(\frac{3}{2}\right) = -3 - \cancel{2} \times \frac{3}{\cancel{2}}$$

$$= -3 - 3$$

$$= -6$$

R.H.L

$$\lim_{x \to \frac{3}{2}^+} f(x) = \lim_{x \to \frac{3}{2}^+} -3 - 2x$$

$$= -3 - \cancel{2} \times \frac{3}{\cancel{2}}$$

$$= -3 - 3$$

$$= -6$$

Since L.H.L = R.H.L, the limit of the function exists.

$$\therefore \lim_{x \to \frac{3}{2}} f(x) = -6$$

Since $\lim\limits_{x \to \frac{3}{2}} \cancel{\neq} f(x) \neq f\left(\frac{3}{2}\right)$, Given function is discontinuous #.

Again,

L.H.L

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} 3+2x$$
$$= 3+2\times 0$$
$$= 3+0$$
$$= 3$$

R.H.L

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} 3-2x$$
$$= 3-2\times 0$$
$$= 3$$

Since L.H.L = R.H.L = 3, the limit of the function exists.

$$\therefore \lim_{x \to 0} f(x) = 3$$

Since $\lim\limits_{x \to 0} f(x) = f(0) = 3$, Given function is continuous #.

4.

(11) a) A function $f(x)$ is defined as

$$f(x) = \begin{cases} 2x^2-5 & \text{for } x>2 \\ 5 & \text{for } x=2 \\ 7-x^2 & \text{for } x<2 \end{cases}$$ Show that

$f(x)$ has removable discontinuity at $x=2$. Also redefined $f(x)$ to make it continuous at $x=2$.

→ Soln,

$f(2)=5$

L.H.L

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} 7-x^2$$

$$= 7-2^2$$

$$= 7-4$$

$$= 3$$

R.H.L

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} 2x^2-5$$

$$= 2\times 2^2-5$$

$$= 8-5$$

$$= 3$$

Since L.H.L = R.H.L = 3, the limit of the function exists.

$$\therefore \lim_{x\to 2} f(x) = 3$$

Since $\lim_{x\to 2} f(x) \neq f(2)$, It has removable discontinuity at $x=2$.

Given function can be make continuous by redefining as.

$$f(x) = \begin{cases} 2x^2-5 & \text{for } x>2 \\ 3 & \text{for } x=2 \\ 7-x^2 & \text{for } x<2 \end{cases}$$

12 b) Given a function $f(x) = \begin{cases} x, & \text{when } 0 \le x < \frac{1}{2} \\ 1, & \text{when } x = \frac{1}{2} \\ 1-x, & \text{when } \frac{1}{2} < x < 1 \end{cases}$

Is the function continuous at $x=\frac{1}{2}$?

If not, redefine $f(x)$ so as to make it continuous at $x=\frac{1}{2}$

→ Soln,

$f\left(\frac{1}{2}\right) = 1$

L.H.L

$$\lim_{x\to \frac{1}{2}^-} f(x) = \lim_{x\to \frac{1}{2}^-} x$$

$$= \frac{1}{2}$$

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R.H.L

$$\lim_{x \to \frac{1}{2}^+} f(x) = \lim_{x \to \frac{1}{2}^+} 1 - x$$

$$= 1 - \frac{1}{2}$$

$$= \frac{2-1}{2}$$

$$= \frac{1}{2}$$

Since, L.H.L = R.H.L = $\frac{1}{2}$, limit of the function exists.

$$\therefore \lim_{x \to \frac{1}{2}} f(x) = \frac{1}{2}$$

Since, $\lim_{x \to \frac{1}{2}} f(x) \neq f\left(\frac{1}{2}\right)$, Given function is discontinuous at $x = \frac{1}{2}$

Given function can be make continuous by redefining as.

$$f(x) = \begin{cases} x; \text{ when } 0 \leq x < \frac{1}{2} \\ \frac{1}{2}, \text{ When } x = \frac{1}{2} \\ 1-x, \text{ when } \frac{1}{2} < x < 1 \end{cases}$$

5.
(13) a) A function $f(x)$ is defined as follows.

$$f(x) = \begin{cases} \dfrac{2x^2-18}{x-3} & \text{for } x \neq 3 \\ K & \text{for } x = 3 \end{cases}$$

Find the value of K so that $f(x)$ is continuous at $x=3$.

→ Soln,
When $x \neq 3$

$$\lim_{x\to 3} f(x) = \lim_{x \to 3} \frac{2x^2-18}{x-3}$$

$$= \cancel{2\times 3^2}\ \frac{2(x^2-9)}{x-3}$$

$$= \frac{2(x+3)\cancel{(x-3)}}{\cancel{(x-3)}}$$

$$= 2(3+3)$$

$$= 2\times 6$$

$$= 12$$

For continuous,

$$L.H.S = R.H.S = \lim_{x\to 3} f(x) = f(3)$$

or, $12 = K$

$\therefore K = 12$ #.

14 b) If $f(x) = \begin{cases} kx-5 & \text{for } x>4 \\ x^2-9 & \text{for } x\le 4 \end{cases}$ is continuous at $x=4$, then find the value of k.

→ Soln,

L.H.L

$$\lim_{x\to 4^-} f(x) = \lim_{x\to 4^-} x^2-9$$

$$= 4^2-9$$

$$= 16-9$$

$$= 7$$

For continuous,

L.H.L = R.H.L

R.H.L

$$\lim_{x\to 4^+} f(x) = \lim_{x\to 4^+} kx-5$$

$$= k\times 4-5$$

$$= 4k-5$$

Now,

L.H.L = R.H.L

or, $7 = 4k-5$

or, $4k = 7+5$

or, $k = \frac{12}{4}$

or, $k = 3$

$\therefore k = 3$

15 (c) Find the value of a in terms of p, if the given function is continuous at the given point $f(x) = \begin{cases} 2ax+3 & \text{if } x<1 \\ 1-px^2 & \text{if } x \geq 1 \end{cases}$ at $x=1$.

→ Soln,

L.H.L

$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} 2ax+3$

$= 2 \times a \times 1 + 3$

$= 2a+3$

R.H.L

$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} 1-Px^2$

$= 1 - P \times 1^2$

$= 1-P$

For continuous,

L.H.L = R.H.L

or, $2a+3 = 1-P$

or, $2a = 1-P-3$

or, $a = \frac{-P-2}{2}$

or, $a = -\left(\frac{P+2}{2}\right)$

$\therefore a = -\left(\frac{P+2}{2}\right)$ #.

6. Find the point(s) of discontinuity of

16 a) $f(x) = \frac{x^2 - 4x + 4}{x - 2}$

→ The function is discontinuous if ~~deter~~ denominator is zero,

i.e. $x - 2 = 0$

or, $x = 2$

$\therefore$ At the point 2, the given expression could not be defined.

So, the function is continuous except 2

~~ ~~ Hence, $f(x)$ is continuous on $(-\infty, 2) \cup (2, \infty)$

17 b) $f(x) = \frac{3x - 1}{x^3 - 5x^2 + 6x}$

→ The function is discontinuous if denominator is zero,

i.e. $x^3 - 5x^2 + 6x = 0$

or, $x(x^2 - 5x + 6) = 0$

or, $x(x^2 - 3x - 2x + 6) = 0$

or, $x[x(x-3) - 2(x-3)] = 0$

or, $x(x-2)(x-3) = 0$

Either,

$x = 0$ $\quad$ $x - 2 = 0$ $\quad$ $x - 3 = 0$

$\quad$ $x = 2$ $\quad$ $x = 3$

$\therefore$ At the point 0, 2 and 3, the given expression could not be defined.

So the function is continuous except 0, 2 and 3,
Hence, $f(x)$ is continuous on $(-\infty,0) \cup (0,2) \cup (2,3) \cup (3,\infty)$.